

V*-fit

VS-fit is a Python application that performs curve fitting using a combination of algebraic and trigonometric polynomials. It also implements periodogram analysis based on the `astropy.timeseries.LombScargle` class¹. The program was inspired by I. L. Andronov's MCV software.

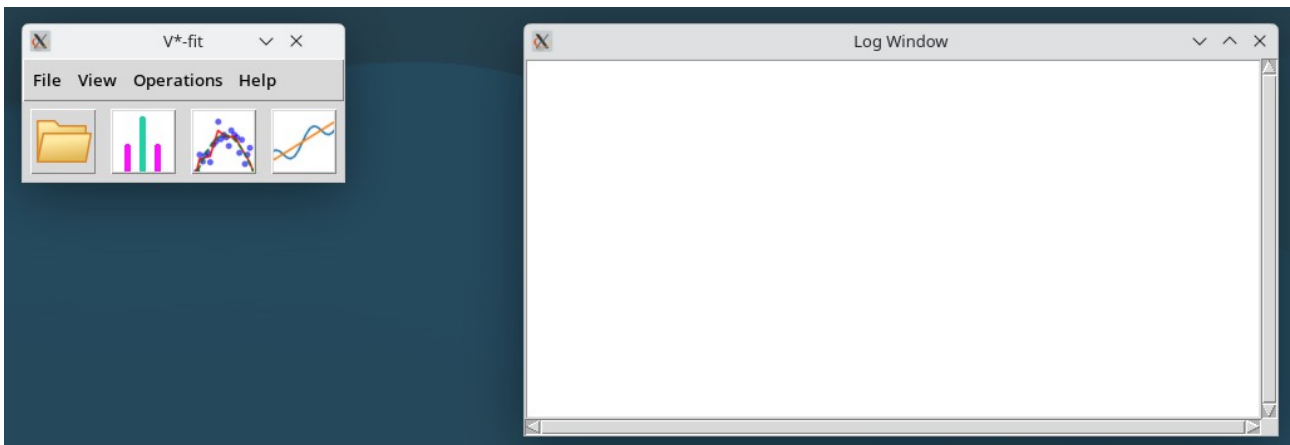
Download the program from its repository at <https://github.com/mpyat2/VS-fit/>. Go to the Releases section (<https://github.com/mpyat2/VS-fit/releases>) and download the ZIP file containing the source code (the program is written in Python, so the source code is essentially the program itself). Then unpack it to any desired location on your machine.

In Windows, launch **VS-fitW.bat**. In Linux, launch **vs-fit.sh**. Note that in both cases it is assumed that Python is installed on the machine. You may edit **VS-fitW.bat** (or **vs-fit.sh**) if the name of the Python interpreter differs from the expected one (**py** in Windows and **python3** in Linux, respectively), or if the interpreter is not in the system PATH.

The program requires the following Python packages:

- astropy
- scipy
- matplotlib
- pandas

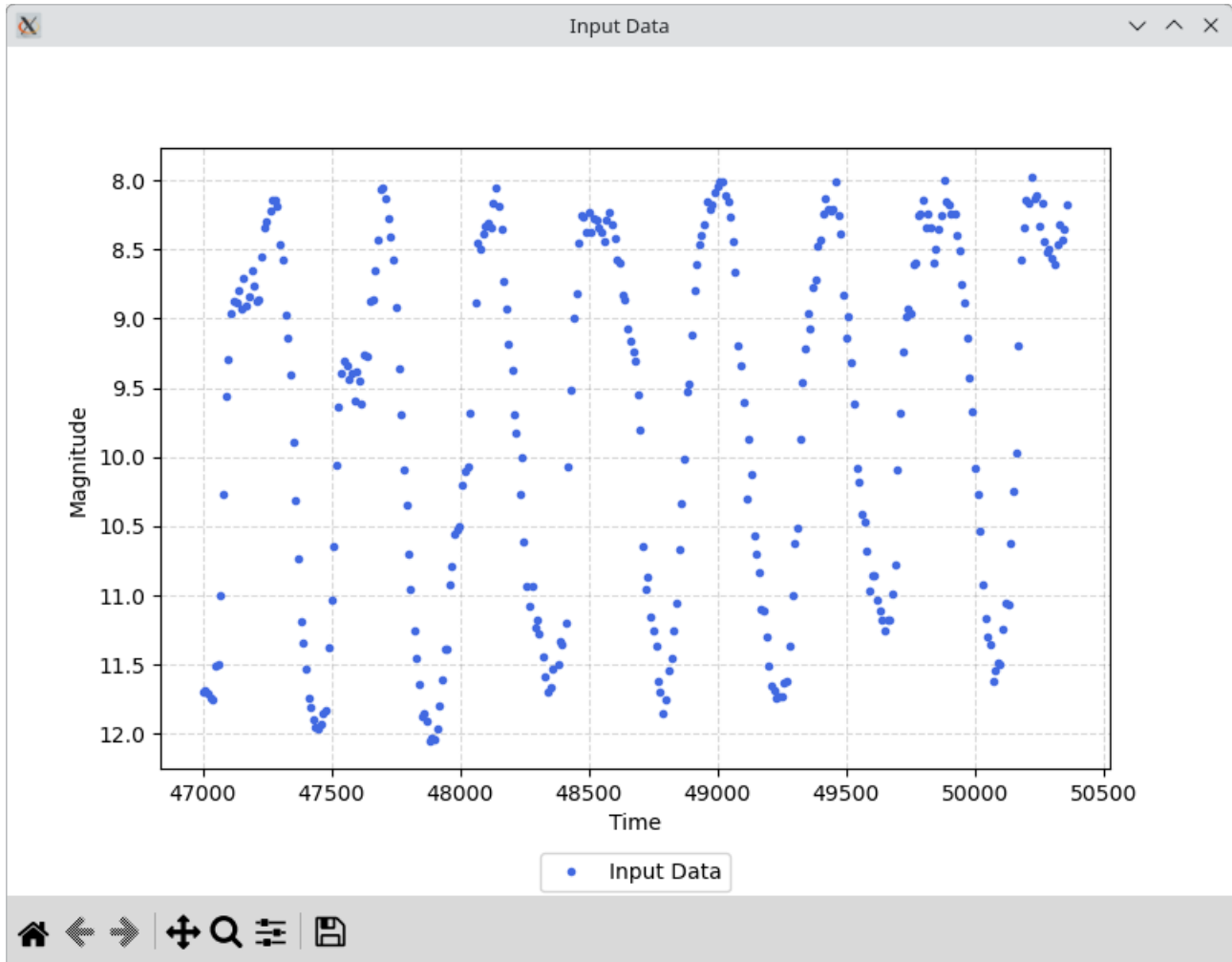
After launching, the main program window and the “Log” window will appear. The main window contains the main menu and toolbar buttons only:



Use the first toolbar button (or select File → Open Data File... from the menu) to open a data file. The program recognizes files with columns delimited by spaces or tabs. Consecutive spaces are treated as a single delimiter, while each tab is treated as a separate delimiter. Empty lines and lines beginning with “#” are ignored. Only the first and second columns are read. It is assumed that the first column contains the X values (i.e., times) and the second column contains the Y values (i.e., magnitudes). Example data files can be found in the **data** folder.

¹ Actually, the name is misleading. The classical Lomb-Scargle method is not even the default option in the Python `LombScargle` class. The default behaviour corresponds to Andronov's periodogram, described in this paper [<https://ui.adsabs.harvard.edu/abs/1994OAP....7...49A/>], in its simplest form.

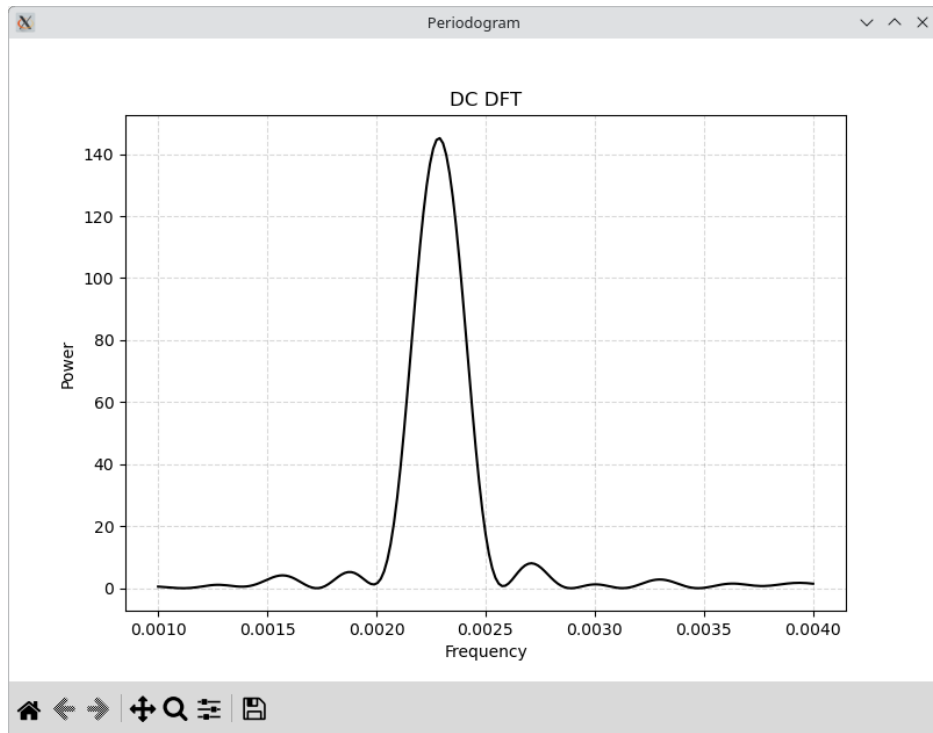
Let's open the **tcas.tsv** file (T Cas observations). A new window will open showing the plot:



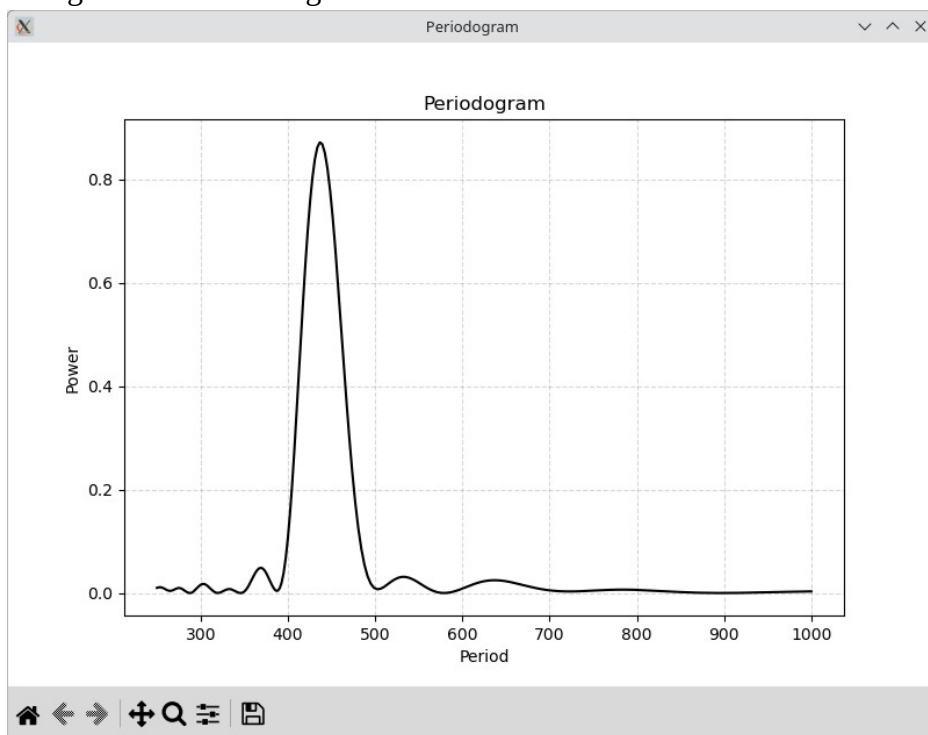
Let's build a periodogram. To do this, select Operations → Periodogram.... Specify the lowest and highest frequencies to analyse, as well as the number of intervals (which is one less than the number of frequencies). For this example, a reasonable choice would be:

The figure is a "Parameters" dialog box. It has four input fields: "Low frequency" with the value 0.001, "High frequency" with the value 0.004, "Intervals" with the value 200, and "Harmonics" with the value 1. There is a question mark icon next to the "Intervals" field. At the bottom, there are "OK" and "Cancel" buttons.

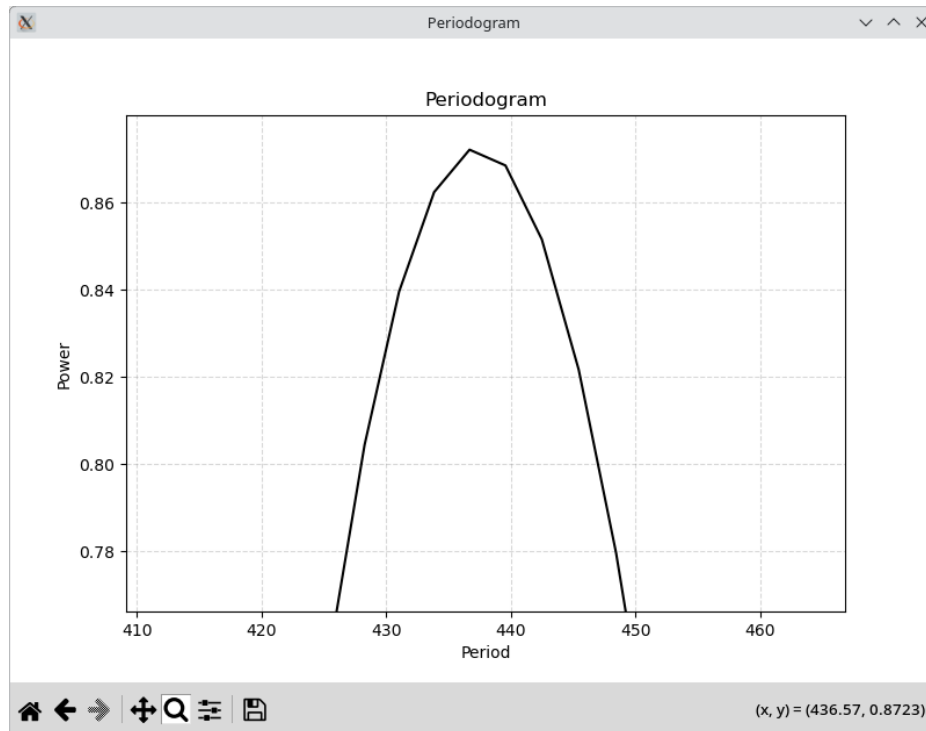
This corresponds to periods ranging from 250 to 1000 days. Press OK. Shortly thereafter, a new window displaying the periodogram will open:



By default, the periodogram is plotted in the frequency domain. You can switch to the period domain by selecting View → Periodogram Power vs Period from the main menu.



Now you can use the magnifying glass icon to zoom in on the region near the peak:



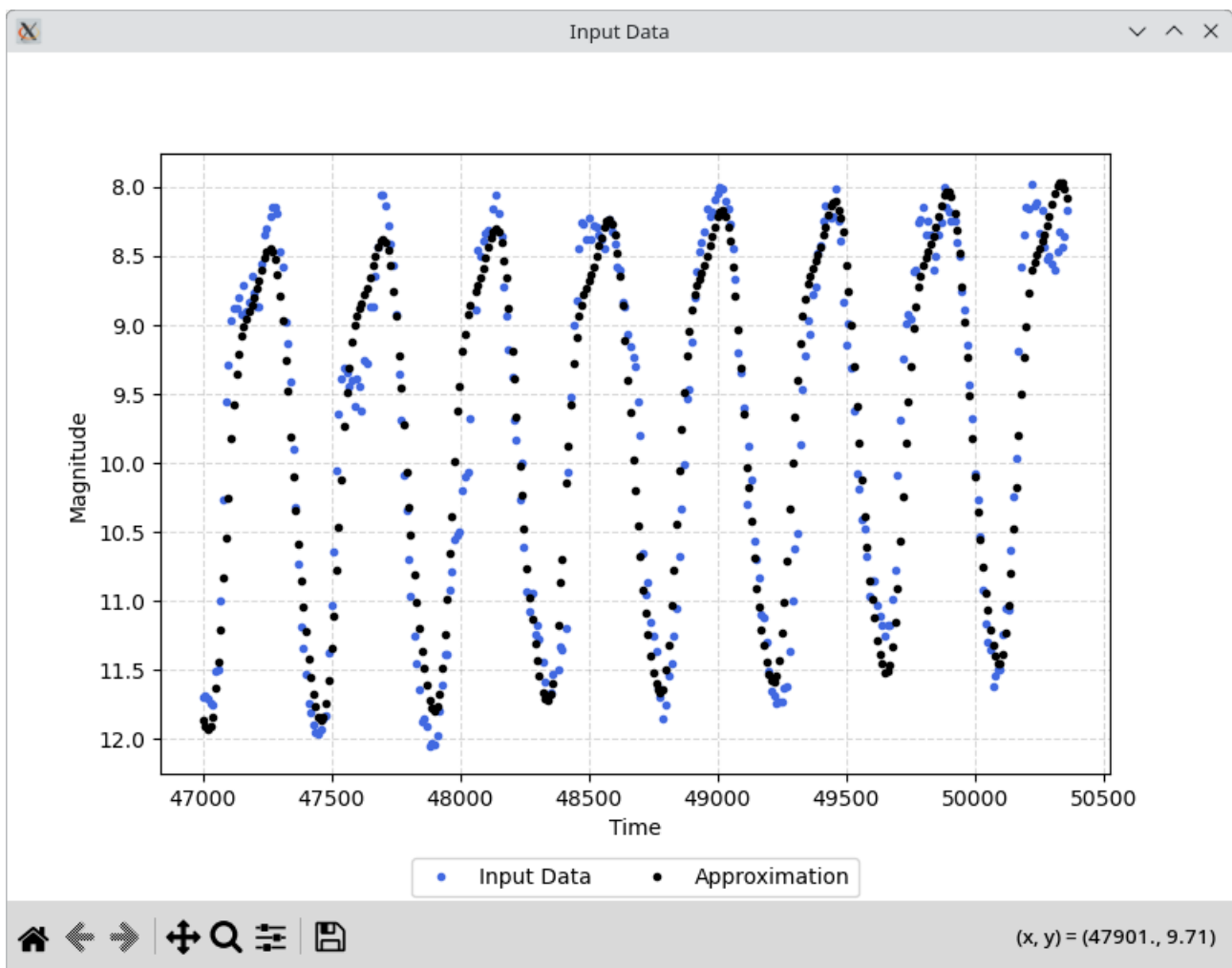
The maximum occurs near a period of approximately 436.6 days. Right-click the maximum to copy its X value (period) to the clipboard. To refine this estimate, select Operations → Polynomial Fit.... You may enter the preliminary period value manually (436.6) or paste it from the clipboard using Ctrl+V. Then set the trigonometric polynomial degree to 4 (an arbitrary choice for demonstration purposes). Also set the algebraic polynomial degree to 1, which is appropriate given the upward trend in the data. Finally, enable the “Optimize” option.

The figure shows a window titled "Parameters" with various input fields and checkboxes. The fields are:

Parameter	Value	Optimize
Polynomial Degree	1	
Trig. Polyn. Period 1	436.6	☒
Trig. Polyn. Period 2	0	☐
Trig. Polyn. Period 3	0	☐
Trig. Polyn. Period 4	0	☐
Trig. Polyn. Period 5	0	☐
Trig. Polyn. Period 6	0	☐
Trig. Polyn. Period 7	0	☐
Trig. Polyn. Period 8	0	☐
Trig. Polyn. Period 9	0	☐

At the bottom, there are checkboxes for "Calculate period errors via bootstrap" (unchecked) and "May take a while!" (unchecked). There are also "OK" and "Cancel" buttons.

Press OK. Shortly, the approximation will be shown in the ‘Input’ window alongside the original data:



Check the Log window. Here you will find the refined period value: 438.97 ± 0.54 days.

```
Log Window

PolyFit started.

=== INITIAL PERIODS ===
period_1 = 436.6000000000000
=====

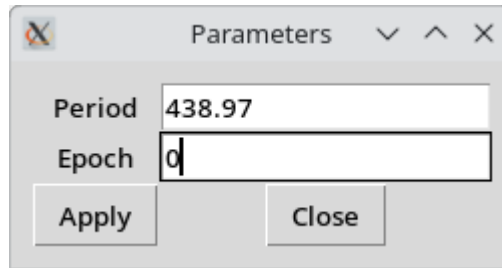
=== OPTIMIZED PERIODS ===
period_1 = 438.973709177896 ± 0.537531072313 (1σ)
=====

=== LINEAR COEFFICIENTS ===
poly_0      : 9.74368832e+00 ± 2.176e-02
poly_1      : -1.54947910e-04 ± 2.235e-05
cos_1       : 5.91421705e-01 ± 3.104e-02
sin_1       : 1.59441244e+00 ± 3.056e-02
cos_2       : -1.63687969e-01 ± 3.066e-02
sin_2       : 2.26644193e-01 ± 3.090e-02
cos_3       : 1.08010336e-01 ± 3.068e-02
sin_3       : -9.60732601e-02 ± 3.089e-02
cos_4       : 7.26619295e-03 ± 3.078e-02
sin_4       : 1.89386131e-02 ± 3.075e-02
=====

=== AMPLITUDES ===
Ampl_1      : 1.70056775e+00 ± 3.059e-02   Period: (1/1) * 438.973709177896
Ampl_2      : 2.79573499e-01 ± 3.077e-02   Period: (1/2) * 438.973709177896
Ampl_3      : 1.44555540e-01 ± 3.072e-02   Period: (1/3) * 438.973709177896
Ampl_4      : 2.02846895e-02 ± 3.081e-02   Period: (1/4) * 438.973709177896
=====

RSS [Sum(O-C)^2] = 5.125016e+01, sigma = 3.977179e-01
Npoints = 335, dof = 324, Nparams = 11
R.M.S. accuracy of the fit sigma[x_c] = 0.07206908805981614
=====
```

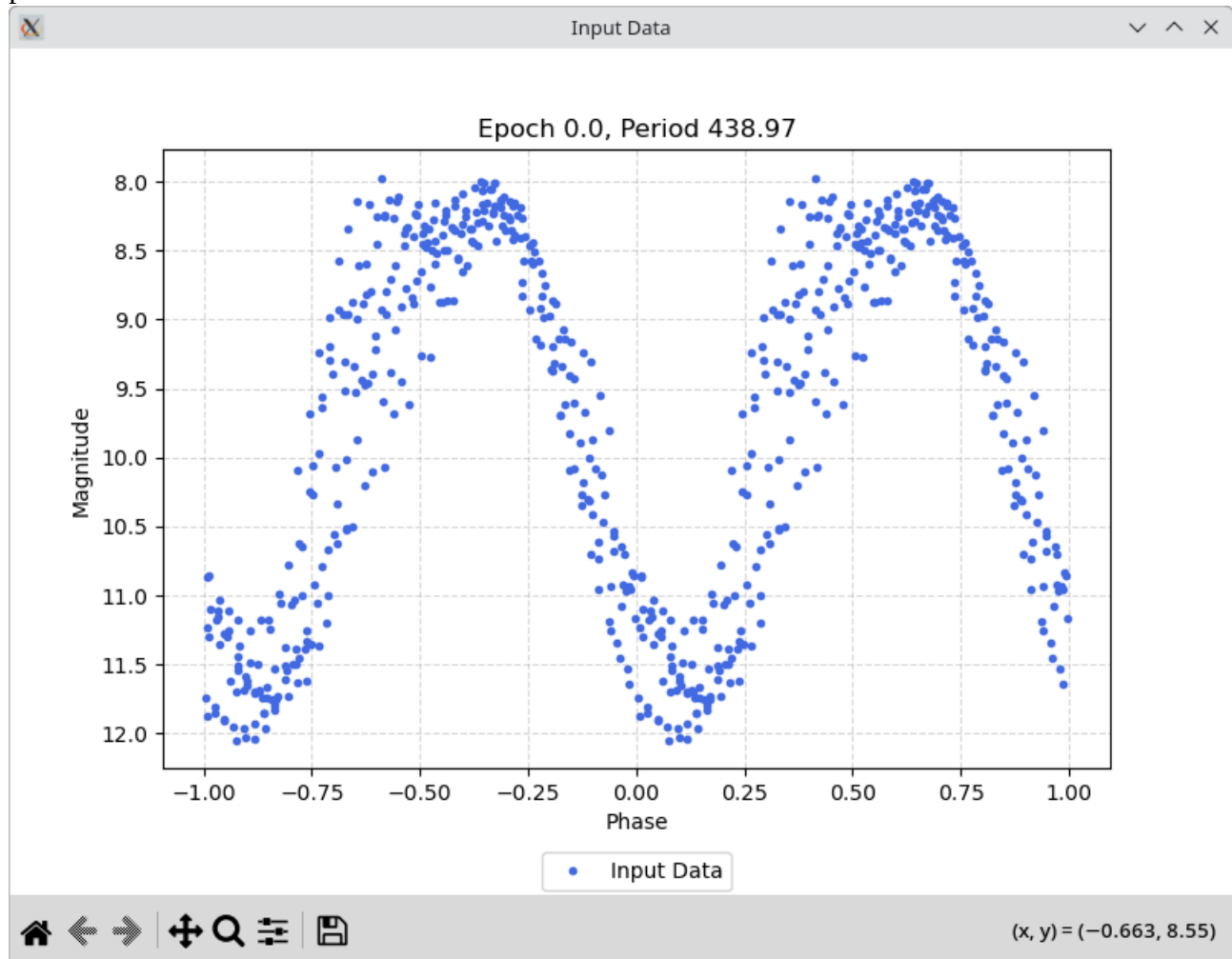
Now, reopen the **tcas.tsv** file, go to the View menu, Phase Plot, and enter the period value:



A dialog box titled "Parameters" with a close button (X) in the top right. It contains two input fields: "Period" with the value "438.97" and "Epoch" with the value "0". Below the fields are two buttons: "Apply" and "Close".

Enter 0 in the “Epoch” field for now, then press “Apply.”

A phase plot will be displayed. You can further refine the initial epoch to align the maximum with phase = 0.



The period value of 438.97 days is quite close to the one listed in the VSX database.

Epoch	16 Nov 2020 (HJD 2459170.0)
Outburst	--
Period	443.5 d
Rise/eclipse dur.	55% (244 d)

(End of the document)